

Part 2 - OOP

Assignment 12 - [Template - Static Members - Exception Handling]

1. Function Template for Finding Maximum

- Create a function template that takes two parameters of any data type.
- The function should return the maximum of the two values.
- Call this function inside the ``main()'` function and test it with different data types.

2. Function Template for Swapping Values

- Create a function template that takes two parameters of any data type by reference.
- The function should swap the values of the two parameters.
- Call this function inside the ``main()'` function and test it with different data types.

3. Static Methods in a Class Template (Calculator Class)

- Create a class template named ``Calculator'` that contains four static methods:
 1. ``add()'` – Takes two parameters of the same data type and returns their sum.
 2. ``subtract()'` – Takes two parameters of the same data type and returns their difference.
 3. ``multiply()'` – Takes two parameters of the same data type and returns their product.
 4. ``divide()'` – Takes two parameters of the same data type and returns their quotient.
- Call each method inside the ``main()'` function.

4. Exception Handling in Division

- Create a ``divide'` function that takes two integers, `x` and `y`, and returns the result of `x / y`.
- Implement exception handling to manage the following cases:
 - If `y = 0`, throw a user-defined exception indicating division by zero is not allowed.
 - If `y < 0`, throw another user-defined exception indicating negative division is not allowed.
- Call this function in ``main()'` and handle the exceptions properly.